

MA7442_A1T1_259039226

Overview

The data set chosen contains 268255 records of information about insurance claims for vehicle repairs. It includes both vehicle specifications including the maker, year registered, miles ran, etc, as well as technical repair data about the vehicle such as the repair cost, hours it will take and repair complexity just to mention a few. Claims fraud is a common occurrence for insurance companies so they keep note of instances where customers might want to take advantage of claims in a fraudulent manner. The data also includes anomalies present in the claims records and categories them into different sectors.

The data set is unique because it captures the economic lifecycle of a claim from the initial vehicle value (Price) to the final repair outcome and whether that outcome was flagged as an anomaly. This allows for a multi-dimensional analysis of financial leakage and fraudulent patterns within the automotive insurance sector.

Aims and Objectives

The key aims are as below:

Aim 1: Assess data quality and integrity

- Identify and standardise missing values or inconsistent entries such as empty strings or variations to ensure a clean foundation for analysis.

Aim 2: Examine feature distributions

- Analyse the spread and skewness of numerical variables to understand the typical range of insurance claims.

Aim 3: Investigate variable relationships

- Explore the correlations between multiple variables to identify cost drivers and claims legitimacy.

Aim 4: Profile anomalies and potential fraud

- Evaluate the frequency and distribution of the anomaly_present flag across different variables to identify high-risk categories.

Aim 5: Analyse temporal trends

- Use certain variables to look for seasonal patterns or duration trends in repair claims over the months and years recorded.

Aim 6: Identify feature suitability for modelling

- Determine which attributes demonstrate the strongest predictive potential or clustering characteristics to inform the subsequent research questions and modelling approaches.

Description of the Data Set

This large-scale dataset facilitates an analysis of the automotive insurance lifecycle particularly concerning financial expenditure and the identification of anomalous insurance claims. The variables are categorised as follows:

Vehicle identity and specification:

Attributes such as Maker, Genmodel, Reg_year, Bodytype and Color establish the baseline characteristics and historical context of the vehicle. These variables are essential for determining the expected valuation and typical wear patterns associated with specific models.

Operational and performance metrics:

Data points including Runned_Miles, Engin_size, Fuel_type and Gearbox allow for a performance-based assessment. These metrics help in understanding how a vehicle's usage and mechanical configuration influence the likelihood and nature of mechanical failures.

Technical repair data:

The core of the claim is represented by the issue, issue_id, repair_complexity and repair_hours. These variables detail the mechanical problems encountered and the technical effort required to return the vehicle to an operational state.

Financial indicators:

Price as the initial vehicle value and repair_cost represent the primary economic variables. Analysing the ratio between these figures is vital for identifying instances of total loss or disproportionate repair spending.

Temporal context:

The inclusion of breakdown_date, repair_date and various Avd time frames allows for the analysis of seasonal trends, repair durations and the age of the claim at the point of processing.

Anomaly and fraud indicators:

The dataset can uniquely detect fraud detection through the anomaly_present and category_anomaly fields. These flags identify claims that deviate from standard patterns providing a ground for predictive modelling and financial leakage analysis.

Exploratory Data Analysis (EDA)

Initial inspection and data profiling

By examining the structure, we identify which variables are categorical and which are continuous, allowing us to plan the appropriate statistical tests and visualisations.

```
# Loading libraries  
library(readr)
```

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ purrr      1.1.0
✓ forcats    1.0.1    ✓ stringr    1.5.2
✓ ggplot2    4.0.0    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
```

```
library(caret)
```

Warning: package 'caret' was built under R version 4.5.2

Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:purrr':

lift

```
library(janitor)
```

Warning: package 'janitor' was built under R version 4.5.2

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(ggplot2)
```

```
library(scales)
```

Warning: package 'scales' was built under R version 4.5.2

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
library(patchwork)
```

Warning: package 'patchwork' was built under R version 4.5.2

```
library(tidyr)
library(tibble)

# Loading the dataset
ds5 <- read_csv("dataset5.csv")
```

Rows: 268255 Columns: 25

— Column specification —————

Delimiter: ","

chr (10): Maker, Genmodel, Genmodel_ID, Color, Bodytype, Gearbox, Fuel_type,...

dbl (15): Adv_year, Adv_month, Reg_year, Runned_Miles, Engin_size, Price, Se...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# A view of the first 10 rows in the data set
head(ds5, 10)
```

A tibble: 10 × 25

	Maker	Genmodel	Genmodel_ID	Adv_year	Adv_month	Color	Reg_year	Bodytype
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	Bentley	Arnage	10_1	2018	4	Silver	2000	Saloon
2	Bentley	Arnage	10_1	2018	6	Grey	2002	Saloon
3	Bentley	Arnage	10_1	2017	11	Blue	2002	Saloon
4	Bentley	Arnage	10_1	2018	4	Green	2003	Saloon
5	Bentley	Arnage	10_1	2017	11	Grey	2003	Saloon
6	Bentley	Arnage	10_1	2017	12	Blue	2002	Saloon
7	Bentley	Arnage	10_1	2018	8	Green	2002	Saloon
8	Bentley	Arnage	10_1	2018	4	Black	2003	Saloon
9	Bentley	Arnage	10_1	2018	2	Silver	2003	Saloon
10	Bentley	Arnage	10_1	2017	11	Silver	2003	Saloon

i 17 more variables: Runned_Miles <dbl>, Engin_size <dbl>, Gearbox <chr>,
 # Fuel_type <chr>, Price <dbl>, Seat_num <dbl>, Door_num <dbl>, issue <chr>,
 # issue_id <dbl>, Adv_day <dbl>, breakdown_date <chr>,
 # repair_complexity <dbl>, repair_cost <dbl>, repair_hours <dbl>,
 # anomaly_present <dbl>, category_anomaly <dbl>, repair_date <chr>

```
# Checking the structure and summary statistics
str(ds5)
```

```
spc_tbl_ [268,255 × 25] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Maker           : chr [1:268255] "Bentley" "Bentley" "Bentley" "Bentley" ...
 $ Genmodel        : chr [1:268255] "Arnage" "Arnage" "Arnage" "Arnage" ...
 $ Genmodel_ID     : chr [1:268255] "10_1" "10_1" "10_1" "10_1" ...
 $ Adv_year        : num [1:268255] 2018 2018 2017 2018 2017 ...
 $ Adv_month       : num [1:268255] 4 6 11 4 11 12 8 4 2 11 ...
 $ Color           : chr [1:268255] "Silver" "Grey" "Blue" "Green" ...
 $ Reg_year        : num [1:268255] 2000 2002 2002 2003 2003 ...
```

```

$ Bodytype      : chr [1:268255] "Saloon" "Saloon" "Saloon" "Saloon" ...
$ Runned_Miles  : num [1:268255] 60000 44000 55000 14000 61652 ...
$ Engin_size    : num [1:268255] 6.8 6.8 6.8 6.8 6.8 6.8 6.8 6.8 6.8 6.8 ...
$ Gearbox       : chr [1:268255] "Automatic" "Automatic" "Automatic" "Automatic" ...
$ Fuel_type     : chr [1:268255] "Petrol" "Petrol" "Petrol" "Petrol" ...
$ Price         : num [1:268255] 21500 28750 29999 34948 26555 ...
$ Seat_num      : num [1:268255] 5 5 5 5 5 5 5 5 4 5 ...
$ Door_num      : num [1:268255] 4 4 4 4 4 4 4 4 4 4 ...
$ issue         : chr [1:268255] "Starter Motor Issue" "Radiator Leaking" "Steering Wheel
Shaking" "Electrical Issue" ...
$ issue_id      : num [1:268255] 0 0 0 4 0 0 0 0 0 0 ...
$ Adv_day       : num [1:268255] 19 15 10 14 6 6 2 3 18 3 ...
$ breakdown_date : chr [1:268255] "19/04/2018" "15/06/2018" "10/11/2017" "14/04/2018" ...
$ repair_complexity: num [1:268255] 3 3 3 3 3 3 3 3 3 3 ...
$ repair_cost    : num [1:268255] 395 695 90 224.8 75.9 ...
$ repair_hours   : num [1:268255] 9 6 3 6 3 1.5 9 9 3 3 ...
$ anomaly_present : num [1:268255] 0 0 0 0 0 0 0 0 0 1 ...
$ category_anomaly : num [1:268255] 0 0 0 0 0 0 0 0 0 0 ...
$ repair_date    : chr [1:268255] "21/04/2018" "16/06/2018" "10/11/2017" "15/04/2018" ...
- attr(*, "spec")=
.. cols(
..   Maker = col_character(),
..   Genmodel = col_character(),
..   Genmodel_ID = col_character(),
..   Adv_year = col_double(),
..   Adv_month = col_double(),
..   Color = col_character(),
..   Reg_year = col_double(),
..   Bodytype = col_character(),
..   Runned_Miles = col_double(),
..   Engin_size = col_double(),
..   Gearbox = col_character(),
..   Fuel_type = col_character(),
..   Price = col_double(),
..   Seat_num = col_double(),
..   Door_num = col_double(),
..   issue = col_character(),
..   issue_id = col_double(),
..   Adv_day = col_double(),
..   breakdown_date = col_character(),
..   repair_complexity = col_double(),
..   repair_cost = col_double(),
..   repair_hours = col_double(),
..   anomaly_present = col_double(),
..   category_anomaly = col_double(),
..   repair_date = col_character()
.. )
- attr(*, "problems")=<externalptr>

```

```
summary(ds5)
```

Maker	Genmodel	Genmodel_ID	Adv_year
Length:268255	Length:268255	Length:268255	Min. :2012

Class :character	Class :character	Class :character	1st Qu.:2018
Mode :character	Mode :character	Mode :character	Median :2018
			Mean :2018
			3rd Qu.:2018
			Max. :2021

Adv_month	Color	Reg_year	Bodytype
Min. : 1.000	Length:268255	Min. :1900	Length:268255
1st Qu.: 4.000	Class :character	1st Qu.:2010	Class :character
Median : 5.000	Mode :character	Median :2014	Mode :character
Mean : 5.626		Mean :2013	
3rd Qu.: 7.000		3rd Qu.:2016	
Max. :33.000		Max. :3010	

Runned_Miles	Engin_size	Gearbox	Fuel_type
Min. : -19966	Min. : 0.100	Length:268255	Length:268255
1st Qu.: 14000	1st Qu.: 1.400	Class :character	Class :character
Median : 39000	Median : 1.800	Mode :character	Mode :character
Mean : 47895	Mean : 9.635		
3rd Qu.: 75000	3rd Qu.: 2.100		
Max. :6363342	Max. :3500.000		

Price	Seat_num	Door_num	issue
Min. : -699925	Min. : 1.000	Min. :0.000	Length:268255
1st Qu.: 4985	1st Qu.: 5.000	1st Qu.:4.000	Class :character
Median : 9289	Median : 5.000	Median :5.000	Mode :character
Mean : 13369	Mean : 5.148	Mean :4.297	
3rd Qu.: 17000	3rd Qu.: 5.000	3rd Qu.:5.000	
Max. :9999999	Max. :20.000	Max. :7.000	

issue_id	Adv_day	breakdown_date	repair_complexity
Min. :0.000	Min. : 1.00	Length:268255	Min. :1.000
1st Qu.:0.000	1st Qu.: 8.00	Class :character	1st Qu.:1.000
Median :0.000	Median :15.00	Mode :character	Median :1.000
Mean :0.499	Mean :14.53		Mean :1.361
3rd Qu.:0.000	3rd Qu.:22.00		3rd Qu.:1.000
Max. :5.000	Max. :28.00		Max. :4.000

repair_cost	repair_hours	anomaly_present	category_anomaly
Min. : -2998.00	Min. : -30.00	Min. :0.0000	Min. :0.00000
1st Qu.: 40.69	1st Qu.: 1.00	1st Qu.:0.0000	1st Qu.:0.00000
Median : 121.50	Median : 2.00	Median :0.0000	Median :0.00000
Mean : 444.61	Mean : 62.07	Mean :0.2115	Mean :0.09308
3rd Qu.: 334.50	3rd Qu.: 6.00	3rd Qu.:0.0000	3rd Qu.:0.00000
Max. :259480.00	Max. :180985.49	Max. :1.0000	Max. :1.00000

```

repair_date
Length:268255
Class :character
Mode :character

```

Data cleaning and pre-processing

We standardise the character strings to ensure that variations in whitespace do not create duplicate categories. Converting the target variable (anomaly_present) and key descriptors into factors is essential for modelling packages. We also remove illogical records such as repairs with zero cost as these likely represent data entry errors or incomplete claims that would skew the final analysis.

```
# Cleaning the data set
ds5_clean <- ds5 %>%
  clean_names() %>%
  mutate(across(where(is.character), ~ trimws(.))) %>%
  mutate(across(where(is.character), ~ na_if(., ""))) %>%
  mutate(
    anomaly_present = factor(anomaly_present, levels = c(0, 1), labels = c("No", "Yes")),
    repair_complexity = as.factor(repair_complexity),
    maker = as.factor(maker),
    fuel_type = as.factor(fuel_type),
    gearbox = as.factor(gearbox)
  ) %>%
  filter(repair_cost > 0) %>%
  select(-genmodel_id, -adv_day) %>% # Dropping low-relevance columns
  drop_na()

# Confirmation of clean data dimensions
dim(ds5_clean)
```

[1] 267110 23

New feature creation

By creating new featured for the data set we transform raw data into domain-specific indicators that reflect real-world insurance dynamics moving beyond basic exploratory analysis. These features allow us to identify financial leakage where costs do not align with mechanical logic and provide a clearer signal for detecting anomalous claims.

```
# Creating indicators to identify financial leakage and usage patterns
ds5_final <- ds5_clean %>%
  mutate(
    vehicle_age = adv_year - reg_year,
    labour_rate = repair_cost / pmax(repair_hours, 1), # pmax function used to prevent division by zero
    cost_to_price_ratio = repair_cost / price,
    annual_mileage = runned_miles / pmax(vehicle_age, 1) # pmax function used to treat new cars
  ) %>%
  # Filtering illogical data to ensure robust modelling and analysis
  filter(vehicle_age >= 0, repair_cost > 0, price > 0)

# Previewing the refined feature set
head(ds5_final %>% select(vehicle_age, labour_rate, cost_to_price_ratio, annual_mileage), 10)
```

A tibble: 10 × 4

	vehicle_age	labour_rate	cost_to_price_ratio	annual_mileage
	<dbl>	<dbl>	<dbl>	<dbl>
1	18	43.9	0.0184	3333.
2	16	116.	0.0242	2750
3	15	30.0	0.00300	3667.
4	15	37.5	0.00643	933.
5	14	25.3	0.00286	4404.
6	15	21.7	0.00130	3667.

7	16	187.	0.0560	4188.
8	15	145.	0.0580	6613.
9	15	25.4	0.00282	1836.
10	14	81.7	0.00817	2714.

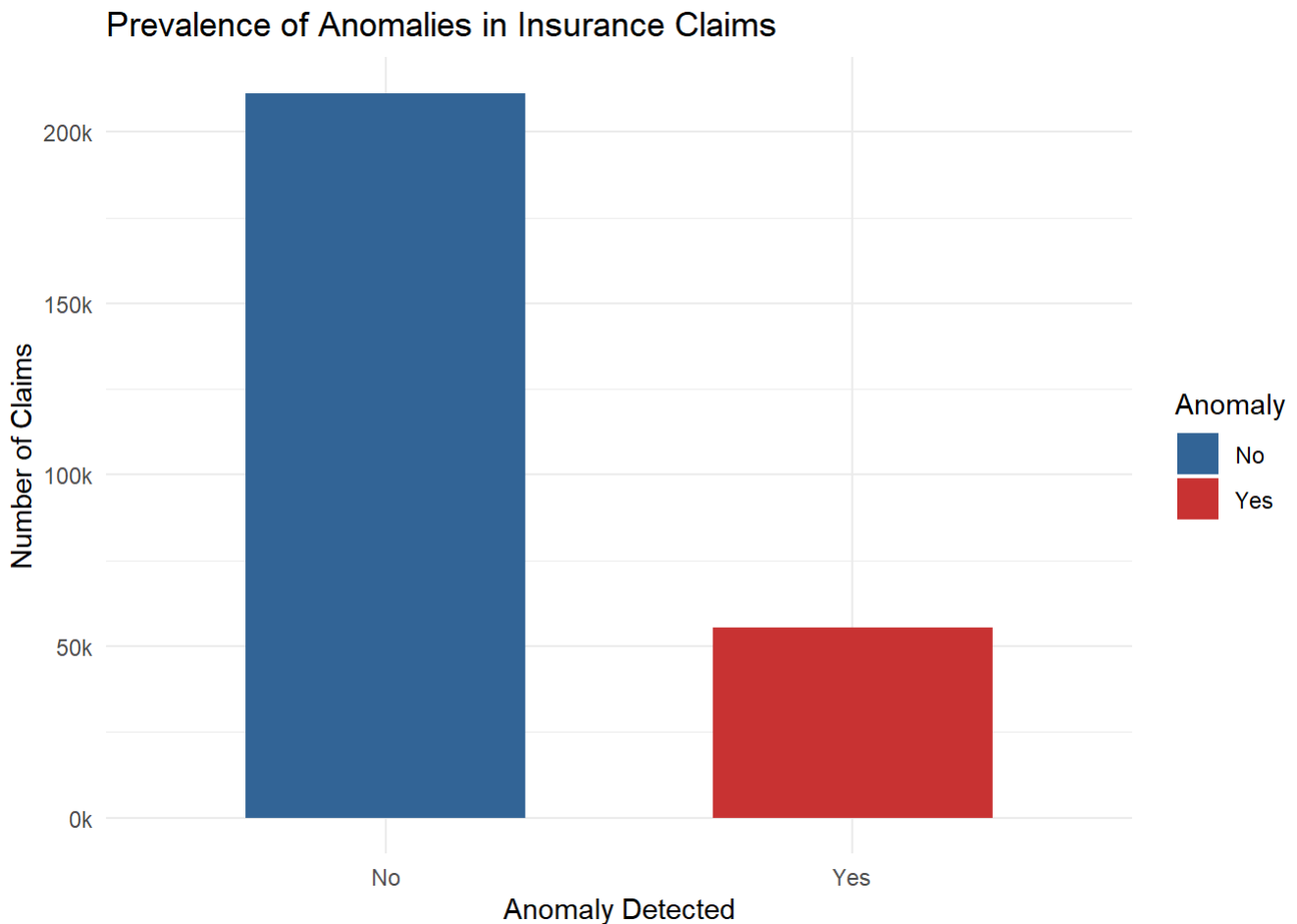
Each of these features serves a specific purpose

- **Vehicle age:** It is essential because the mechanical integrity and the availability of original manufacturer parts vary significantly between a 2-year-old vehicle and a 15-year-old vehicle. Older cars are more prone to wear and tear while anomalies in newer cars might suggest manufacturing defects or deliberate fraud.
- **Labour rate:** Automotive industry garages typically operate within standard hourly rate brackets. If a specific claim demonstrates an exemplary labour_rate of £500/hour while the industry average is closer to £80/hour, it serves as a high probability indicator for billing anomalies and financial leakage.
- **Cost-to-price ratio:** It is used as a standard industry metric for determining total loss. In insurance, if the repair cost exceeds a specific percentage of the car's market value the repair is most often deemed uneconomic and written off. Identifying completed repairs with a very high ratio can highlight inefficiencies in the claims process.
- **Annual mileage (usage intensity):** Total mileage can be misleading without the context of time. By calculating the annualised mileage we can distinguish between a high-use vehicle likely commercial or long-distance and a low-use vehicle which significantly impacts the wear and tear expectations for an insurance claim.

Visualisations and establishing relationships

Prevalence of anomalies in insurance claims

```
# Visualising the prevalence of anomalies
ggplot(ds5_final, aes(x = anomaly_present, fill = anomaly_present)) +
  geom_bar(width = 0.6) +
  scale_fill_manual(values = c("#336699", "#CC3333")) +
  theme_minimal() +
  labs(title = "Prevalence of Anomalies in Insurance Claims",
       x = "Anomaly Detected",
       y = "Number of Claims",
       fill = "Anomaly") +
  scale_y_continuous(labels = label_number(suffix = "k", scale = 1e-3))
```



- Verdict: Anomalous claims are rare events creating a significant class imbalance.

The chart shows that legitimate claims (blue) are the vast majority, while anomalies (red) are much less frequent. This tells us that an anomaly is deviation from the norm rather than a common occurrence. For our future modelling this means that we must focus on how to accurately catch these rare 'red flags' without getting distracted by the sheer volume of normal data.

Proportion of anomalies by vehicle maker

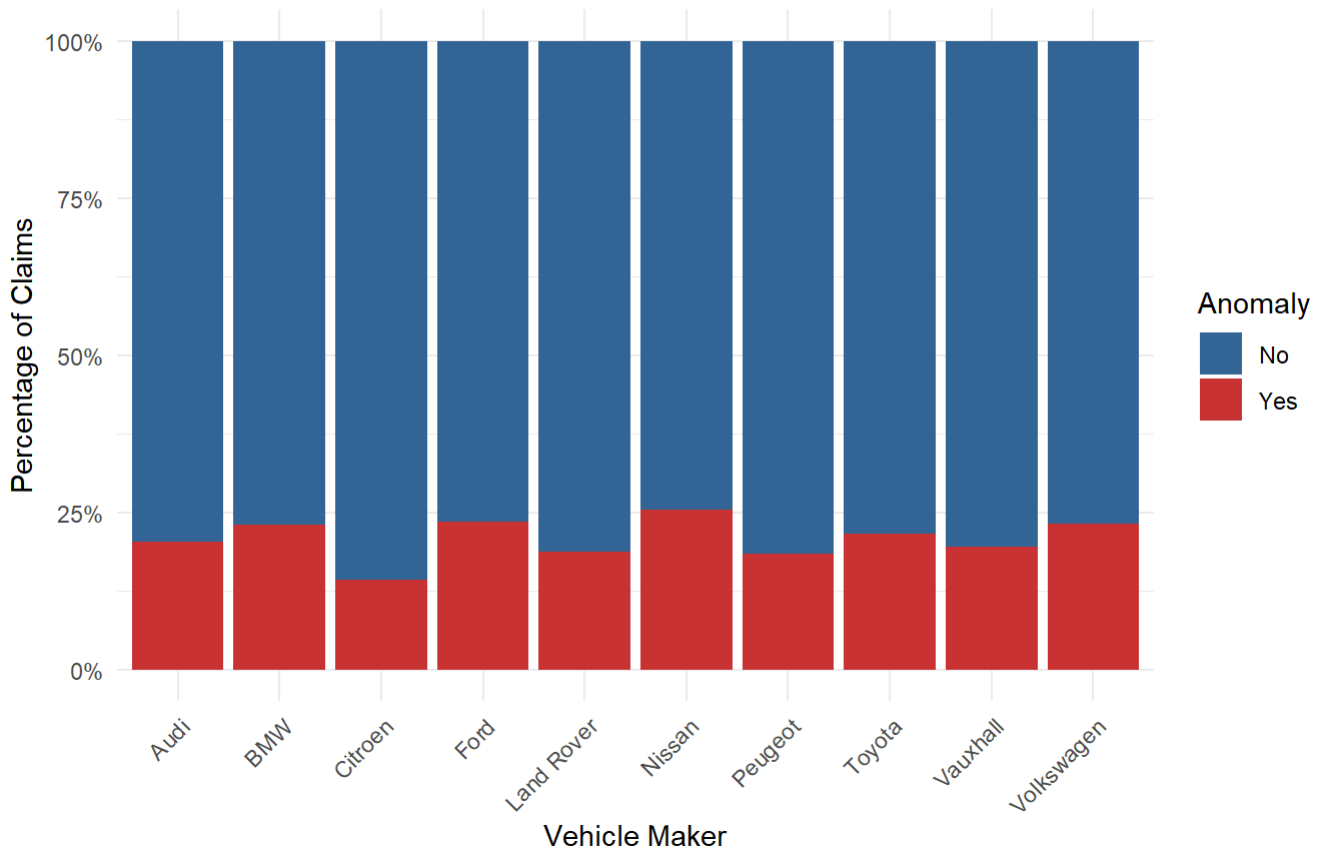
```
# Identifying the top 10 makers by claim volume
top_makers <- ds5_final %>%
  group_by(maker) %>%
  summarise(count = n()) %>%
  arrange(desc(count)) %>%
  slice(1:10) %>%
  pull(maker)

# Plot the anomaly split for these makers
ds5_final %>%
  filter(maker %in% top_makers) %>%
  ggplot(aes(x = maker, fill = anomaly_present)) +
  geom_bar(position = "fill") + # Shows the percentage/proportion
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("#336699", "#CC3333")) +
  theme_minimal() +
  labs(title = "Proportion of Anomalies by Vehicle Maker",
       subtitle = "Comparing the anomaly rate across the top 10 most frequent manufacturers.",
```

```
x = "Vehicle Maker",
y = "Percentage of Claims",
fill = "Anomaly") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Proportion of Anomalies by Vehicle Maker

Comparing the anomaly rate across the top 10 most frequent manufacturers.



- Verdict: The risk of a claim being flagged as an anomaly varies across different car manufacturers.

By looking at the percentage rather than just the total number of cars we can see if certain brands have a higher rate of suspicious claims. If the red section is larger for one brand, it suggests that manufacturer's repair costs or parts might be more prone to inconsistencies justifying a deeper look into brand-specific risk assessment.

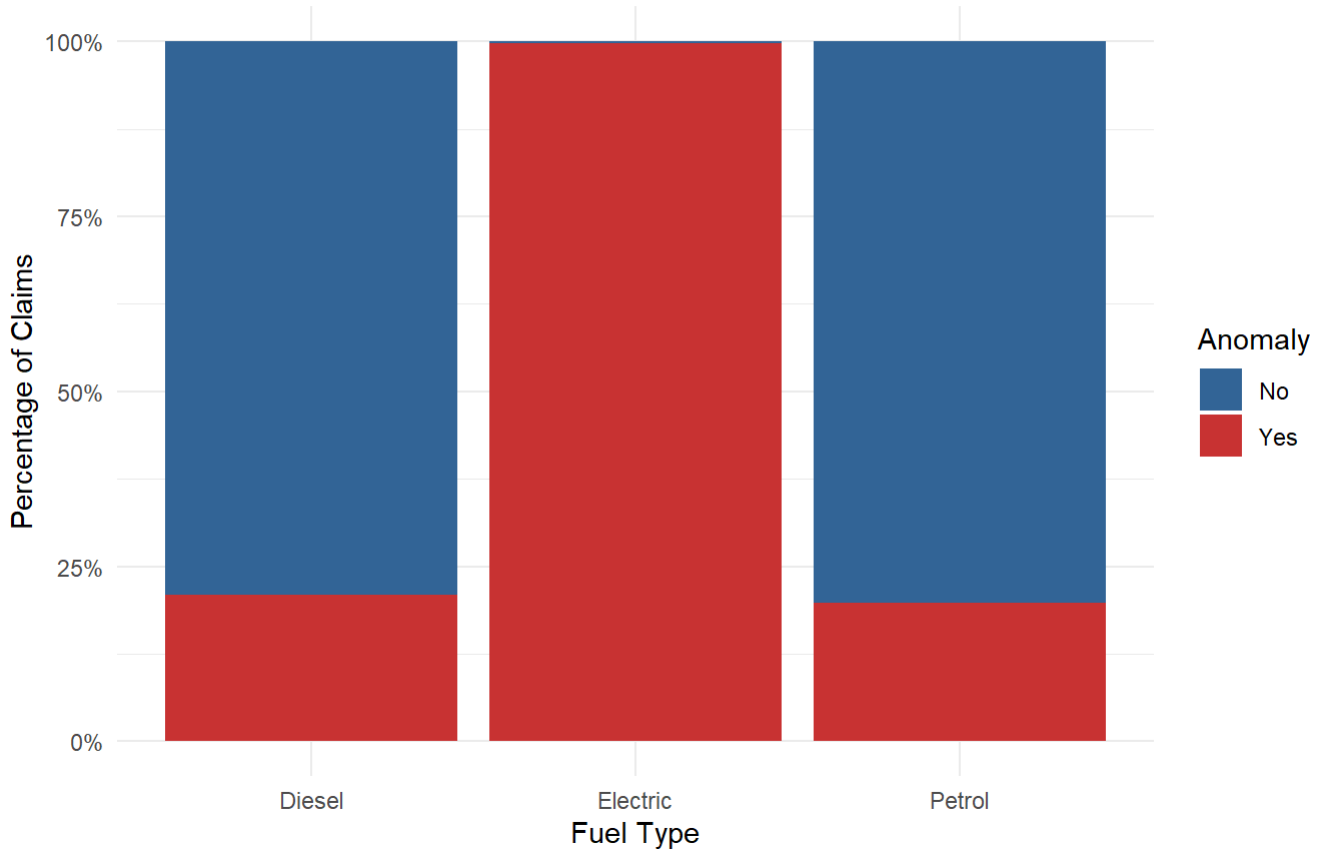
Fuel type and anomaly probability

```
# Fuel Type vs Anomaly Presence
ds5_final %>%
  # Filtering for the most common fuel types to avoid cluttering the plot
  filter(fuel_type %in% c("Petrol", "Diesel", "Hybrid", "Electric")) %>%
  ggplot(aes(x = fuel_type, fill = anomaly_present)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("#336699", "#CC3333")) +
  theme_minimal() +
  labs(title = "Proportion of Anomalies by Fuel Type",
       subtitle = "Evaluating if engine technology impacts the integrity of insurance claims.",
       x = "Fuel Type",
```

```
y = "Percentage of Claims",
fill = "Anomaly")
```

Proportion of Anomalies by Fuel Type

Evaluating if engine technology impacts the integrity of insurance claims.



- Verdict: Engine technology shows a stable anomaly rate suggesting fraud is driven by billing rather than powertrain composition.

When looking at the bars for Petrol and Diesel, the red portion (anomalies) remains remarkably consistent at roughly 20% across the board. The bar for Electric powertrains however indicate close 100% anomalies in our data which could indicate an error to be investigated in the flagging system. If the data was to be excluded, it could reinforce an earlier finding that the unusual claims are likely found in the financial data like labour rates rather than the physical attributes of the car like colour.

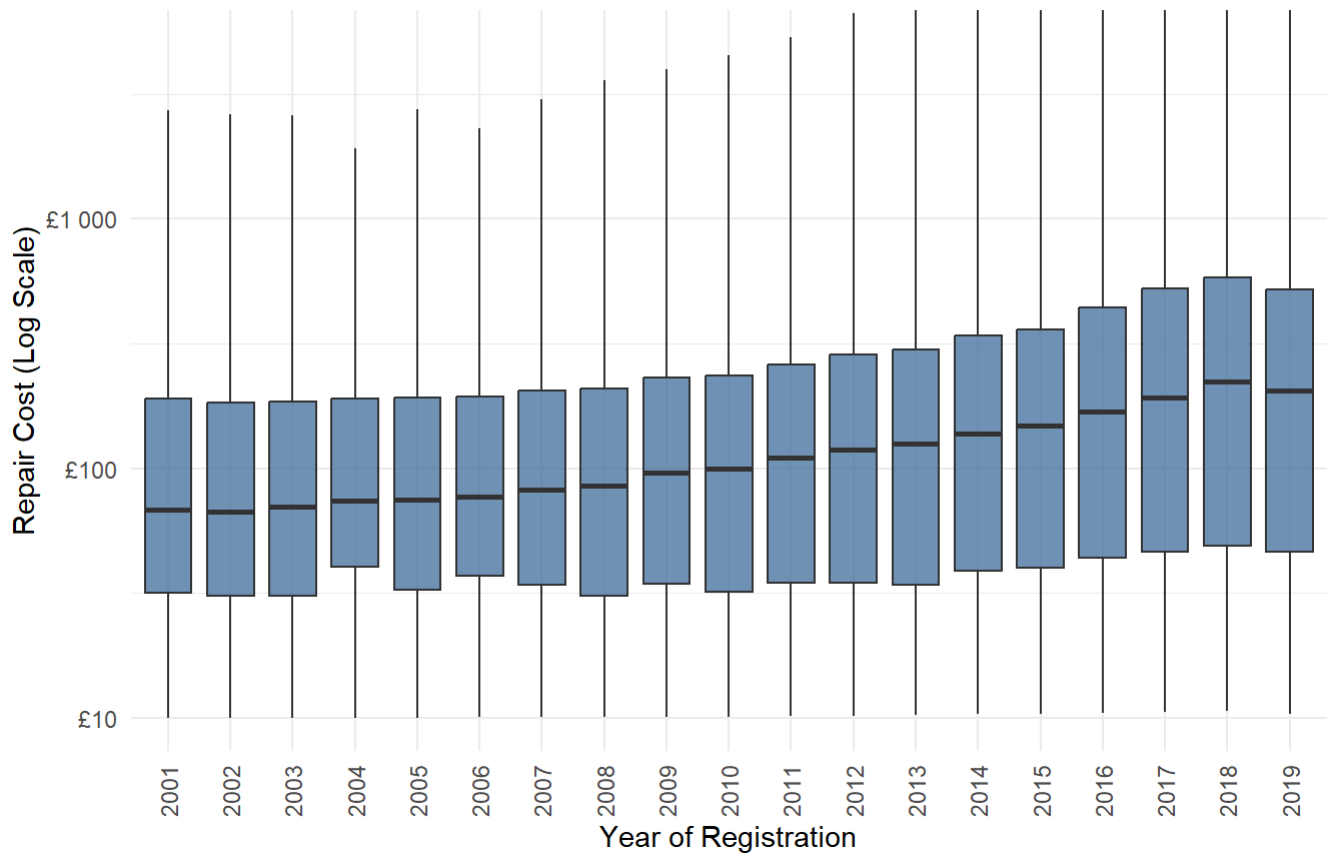
Vehicle registration year vs repair cost

```
# Registration Year vs Repair Cost using a boxplot grouped by year to see the trend over time
ds5_final %>%
  filter(reg_year > 2000) %>% # Focus on more recent history for clarity
  ggplot(aes(x = as.factor(reg_year), y = repair_cost)) +
  geom_boxplot(outlier.shape = NA, fill = "#336699", alpha = 0.7) +
  scale_y_log10(labels = label_number(prefix = "£")) +
  coord_cartesian(ylim = c(10, 5000)) + # Focus on the bulk of the data
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5)) +
  labs(title = "Repair Cost Trends by Vehicle Registration Year",
       subtitle = "Observation of how the manufacturing year influences standard repair expend
```

```
x = "Year of Registration",
y = "Repair Cost (Log Scale)"
```

Repair Cost Trends by Vehicle Registration Year

Observation of how the manufacturing year influences standard repair expenditure.



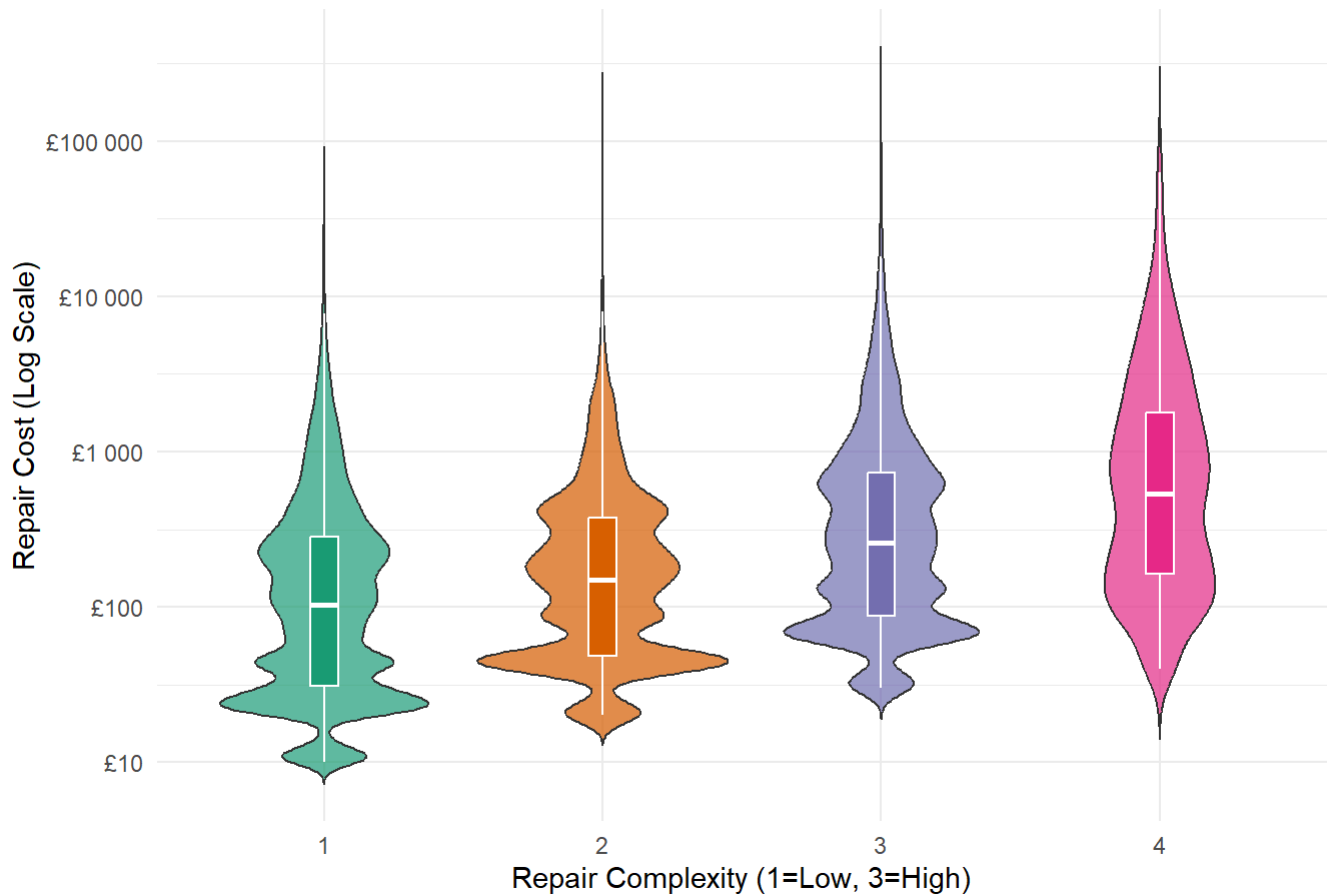
- Verdict: Newer vehicles have generally higher fixed costs than older cars.

Reasoning: The boxplots show a clear upward trend as cars registered in 2018-2020 have higher repair costs, likely due to expensive sensors and modern technology. If you look at cars from the early 2000s, the repair costs are not too far from the new cars. This may mean that while an old car can be cheap to fix it is also prone to sudden extreme repair bills possibly because finding parts for a 20-year-old vehicle can be difficult and expensive.

Repair cost distribution by complexity tier

```
# Examining the consistency of costs across technical difficulty levels.
ggplot(ds5_final, aes(x = repair_complexity, y = repair_cost, fill = repair_complexity)) +
  geom_violin(alpha = 0.7, trim = FALSE) +
  geom_boxplot(width = 0.1, color = "white", outlier.shape = NA) +
  scale_y_log10(labels = label_number(prefix = "£")) +
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() +
  labs(title = "Repair Cost Distribution by Complexity Tier",
       x = "Repair Complexity (1=Low, 3=High)",
       y = "Repair Cost (Log Scale)") +
  theme(legend.position = "none")
```

Repair Cost Distribution by Complexity Tier



- Verdict: Costs generally rise with complexity, but simple repairs can often have suspiciously high price tags.

While complex repairs are usually the most expensive, the low complexity (Level 1) section contains many outliers that cost as much as a major engine rebuild. These specific cases represent financial leakage representing a simple job being billed at an extremely high rate and making complexity a key area for spotting errors.

Correlation heatmap of vehicle and repair metrics

```
# Identifying linear relationships between engineered features and anomalies using a correlat
# Preparation of the numeric data
corr_data <- ds5_final %>%
  select(price, runned_miles, repair_cost, repair_hours,
         vehicle_age, labour_rate, cost_to_price_ratio, annual_mileage) %>%
  mutate(anomaly_flag = as.numeric(ds5_final$anomaly_present) - 1)

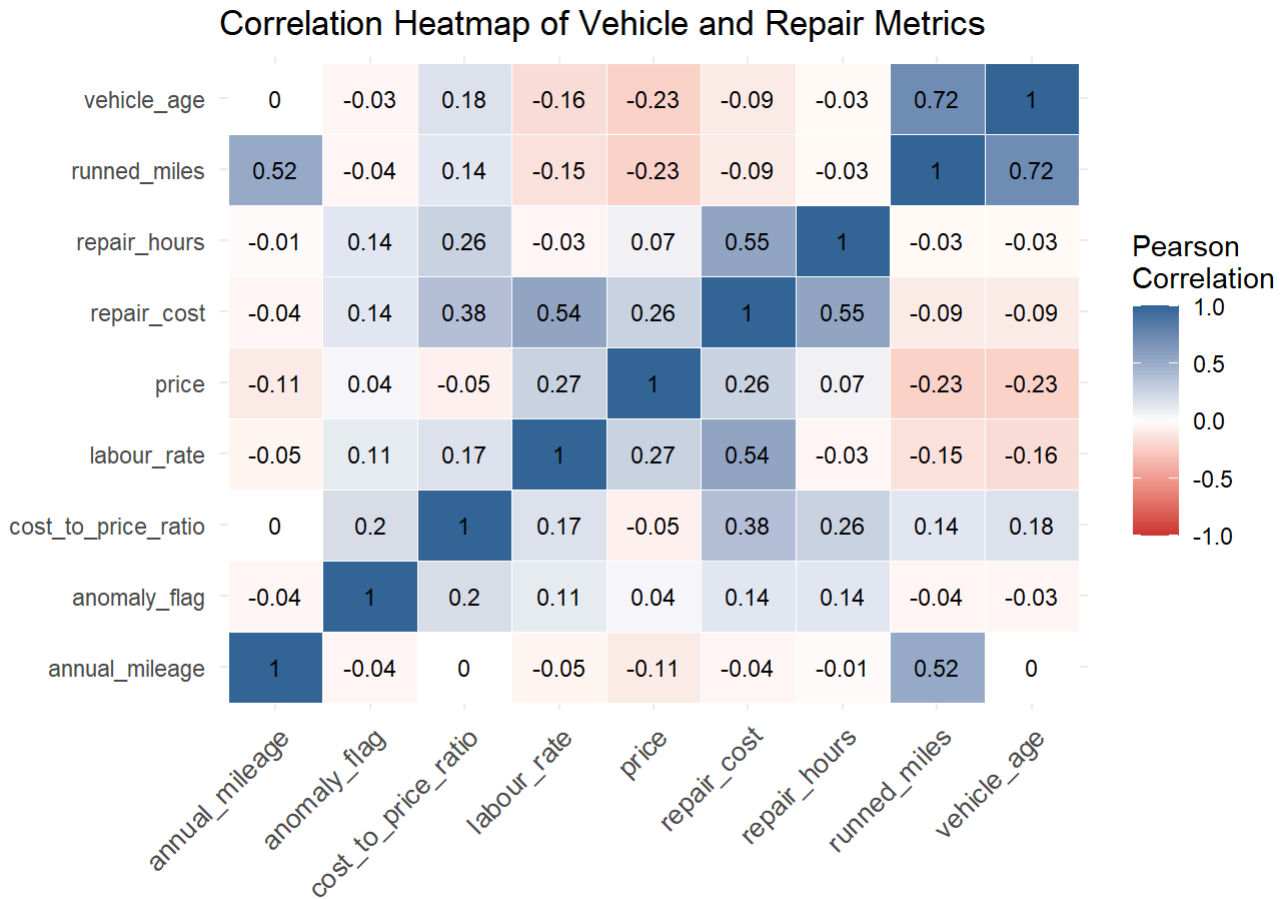
# Computing the correlation and converting to 'long' format for ggplot
corr_matrix <- cor(corr_data, use = "complete.obs")
corr_long <- as.data.frame(corr_matrix) %>%
  rownames_to_column(var = "Variable1") %>%
  pivot_longer(cols = -Variable1, names_to = "Variable2", values_to = "Correlation")

# Plotting using geom_tile
ggplot(corr_long, aes(x = Variable1, y = Variable2, fill = Correlation)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "#CC3333", high = "#336699", mid = "white",
```

```

midpoint = 0, limit = c(-1,1), space = "Lab",
name="Pearson\nCorrelation") +
geom_text(aes(label = round(Correlation, 2)), size = 3) + # Adds values to boxes
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, size = 10, hjust = 1)) +
labs(title = "Correlation Heatmap of Vehicle and Repair Metrics",
x = "", y = "")

```



- Verdict: Engineered features like labour rate and cost-to-price ratio are stronger indicators of anomalies than raw data.

The heatmap shows how variables relate to one another. While most factors behave as expected such as older cars costing less, the cost-to-price ratio has a notable link to the anomaly flag. This confirms that repairs costing a huge chunk of the car's total value are more likely to be flagged as suspicious or uneconomic.

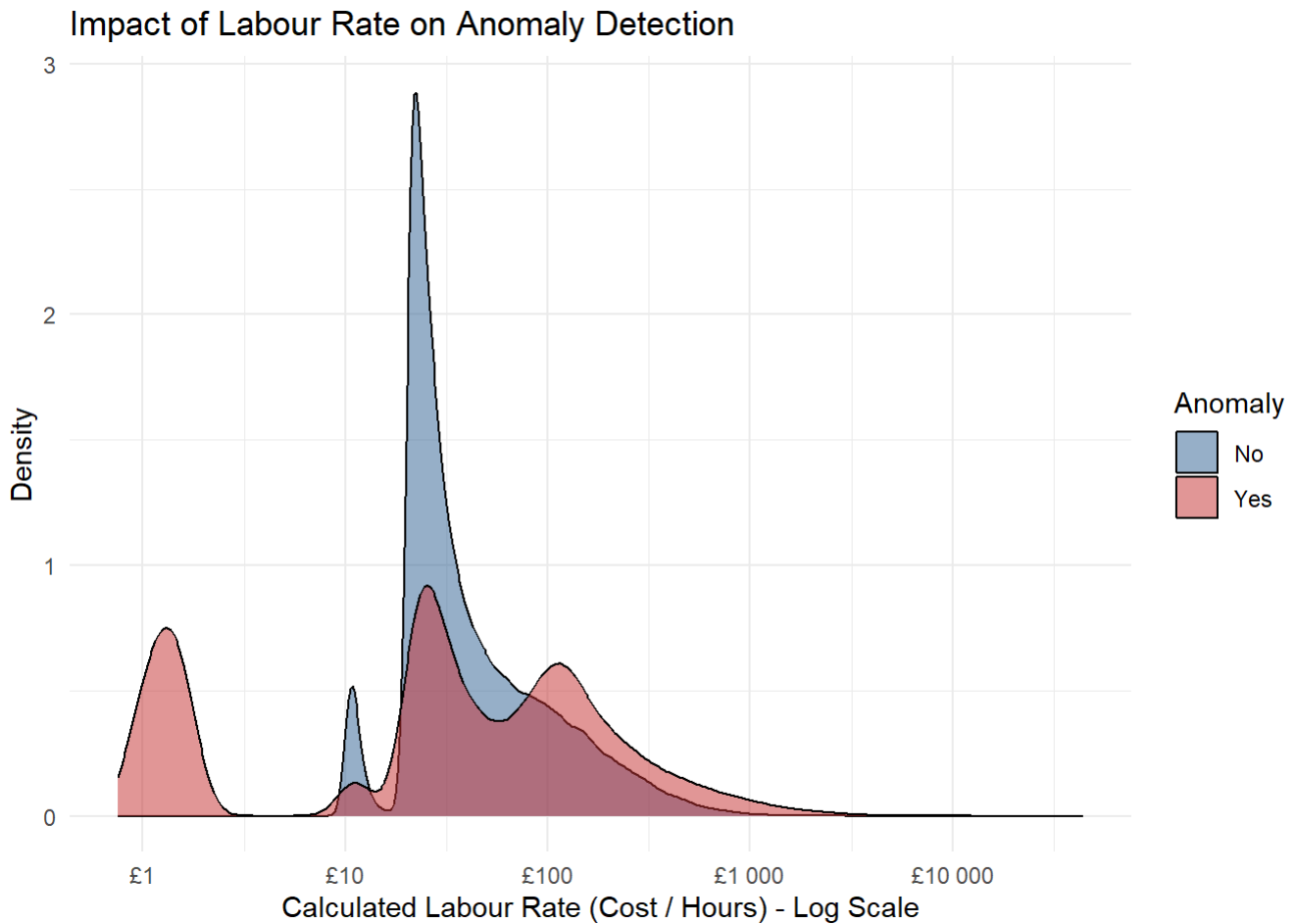
Impact of labour rate on anomaly detection

```

# Density of labour rate by anomaly status
ggplot(ds5_final, aes(x = labour_rate, fill = anomaly_present)) +
  geom_density(alpha = 0.5) +
  scale_x_log10(labels = label_number(prefix = "£")) +
  scale_fill_manual(values = c("#336699", "#CC3333")) +
  theme_minimal() +
  labs(title = "Impact of Labour Rate on Anomaly Detection",

```

```
x = "Calculated Labour Rate (Cost / Hours) - Log Scale",  
y = "Density",  
fill = "Anomaly")
```



- Verdict: Anomalous claims often exhibit irregular hourly billing rates.

This is our most revealing chart. The red bump for anomalous claims is shifted further to the right than the blue normal claims. This proves that high labour rates are a massive red flag meaning that suspicious claims are not just expensive because parts are pricey, but because the time billed is being charged at an unusually high rate.

Developed Research Questions

Question 1: How does the relationship between repair expenditure and vehicle market value (cost-to-price ratio) serve as a predictor for claim anomalies across different vehicle age brackets?

Rationale:

The correlation heat-map established a clear link between the cost-to-price ratio and anomalous outcomes. In the insurance industry, repairs that approach the car's total value are considered uneconomic. This question investigates whether over-repairing a vehicle is a common trait of suspicious claims or a sign of inefficient processing helping to identify where the threshold of bad value usually lies. Santolino et al. [1] demonstrate that vehicle age significantly impacts the expected severity of a

claim showing that outcomes remain constant or worsen as a vehicle passes the 18 year mark. This validates the use of threshold analysis to determine if older vehicles are disproportionately flagged as total losses despite being repairable.

This question also addresses whether total loss thresholds disproportionately affect lower-income individuals who are more likely to own older lower value vehicles. There is an ethical risk that automated models could systematically disadvantage these groups by frequently flagging their claims as uneconomic. The global economic system aimed at eliminating waste by keeping products and materials in use for as long as possible promotes the longevity of assets. The rationale then explores how insurers can balance financial risk with the environmental benefit of keeping older functional vehicles on the road rather than forcing a replacement.

Question 2: To what extent can automated classification models detect invoice padding by identifying claims where the hourly labour rate significantly exceeds industry standards?

Rationale:

The density distribution analysis showed that anomalous claims frequently occupy a distinct high-cost tail in the labour rate distribution suggesting inflated billing. This aligns with modern insurance challenges where costs for complex vehicles have risen by 65% due to rising labour and part costs. Anand Kumar and Sountharajan [2] propose a framework for accurate claims estimation using advanced neural networks providing that automated systems can distinguish legitimate cost increases from claim padding.

Insurance providers must ensure that fraud detection does not become a barrier for the customers' right to repair. If algorithms are too aggressive in flagging high labour rates they may unfairly penalise independent, local garages that lack the financial muscle of large chains. From a sustainability perspective it is vital that these models do not unfairly encourage writing off vehicles that could be safely repaired as premature scrapping increases mechanical waste and the carbon footprint associated with manufacturing new vehicles.

Question 3: Can we discover hidden groups of claims that show similar patterns of high mileage and high repair frequency to identify commercial vehicles being used for non-disclosed purposes?

Rationale:

The new annual_mileage feature highlights vehicles with usage intensity far beyond the average for their age. High-mileage vehicles that frequently undergo complex repairs may indicate that the car is being used for commercial work not disclosed on the insurance policy such as courier work or unlicensed taxis. Unsupervised learning models such as K-Means clustering allows for the identification of groups of high-risk users based on their actual driving behaviour rather than just their stated policy details. Hainaut [3] demonstrates how clustering can allow for identification of latent structures of policyholders by risk behaviours. This allows insurance firms to identify groups of users who share similar usage patterns, helping to separate genuine high-intensity drivers from those misusing their policies.

From an ethical scope, insurers must also distinguish between policy fraud and geographical necessity. Individuals in rural areas with poor public transport infrastructure are forced to have higher annual

mileages. A model that flags high mileage as suspicious without context could unfairly penalise rural or marginalised communities who have no choice but to drive long distances for work. The goal is to ensure the model identifies commercial exploitation rather than punishing essential travel.

Aims and Objectives

Aim:

- To develop a data-driven framework for identifying financial leakage and fraudulent patterns within the automotive insurance claims sector.

Objectives:

- To profile risk by conducting a multi-dimensional analysis of vehicle metadata such as maker and fuel type to identify high-risk segments for insurance providers.
- To predict anomalies by developing and evaluating supervised classification models like logistic regression and advanced neural networks to automate the detection of fraudulent claims.
- To uncover hidden patterns through utilising clustering techniques, for example K-means, to group claims into economic tiers identifying hidden patterns where repair costs disproportionately exceed vehicle value.

References

- [1] M. Santolino, L. Céspedes, and M. Ayuso, "The Impact of Aging Drivers and Vehicles on the Injury Severity of Crash Victims," *International Journal of Environmental Research and Public Health*, vol. 19, no. 24, p. 17097, Dec. 2022. DOI: 10.3390/ijerph192417097.
- [2] P. Anand Kumar and S. Sountharajan, "Insurance claims estimation and fraud detection with optimized deep learning techniques," *Scientific Reports*, vol. 15, no. 1, p. 27296, Jul. 2025. DOI: 10.1038/s41598-025-12848-0.
- [3] D. Hainaut, "Insurance Analytics with Clustering Techniques," *Risks*, vol. 12, no. 9, p. 141, 2024. DOI: 10.3390/risks12090141.